

ADAPTIVE STRUCTURE-AWARE REPULSION: GUIDING GRAPH CONTRASTIVE LEARNING BEYOND NEURAL COLLAPSE

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ABSTRACT

Graph Contrastive Learning (GCL) has emerged as a powerful paradigm for self-supervised representation learning on graphs. However, its theoretical underpinnings are at a turning point. The prevailing Neural Collapse theory, which posits a Simplex Equiangular Tight Frame (ETF) as the optimal geometry, is being challenged by a more nuanced ‘low-rank bias’ theory, suggesting that deep models favor more structured, lower-rank solutions. This exposes a critical gap: existing GCL methods are not designed for this new paradigm. This is a challenging problem because current regularizers are either computationally prohibitive for large graphs, scaling quadratically with the number of nodes, or they are inefficient, applying indiscriminate repulsive forces that waste model capacity on trivial negative samples. To bridge this gap, we introduce Adaptive Structure-Aware Repulsion (ASAR), a novel and scalable loss-based regularizer that explicitly operationalizes the low-rank bias theory. ASAR features a dual-adaptive mechanism: it dynamically identifies a set of ‘confusable’ hard negatives for each anchor based on an adaptive similarity margin, and it applies a localized repulsive force whose strength is modulated by a self-organizing feedback controller that responds to the batch’s representation geometry. This makes ASAR a unique, local, and adaptive regularizer with a computational complexity of $O(b^2)$, making it suitable for massive graphs. We designed a comprehensive experimental protocol to verify ASAR’s efficacy on large-scale OGB benchmarks, aiming to demonstrate its superiority over standard GCL baselines and validate its theoretical contributions.

1 INTRODUCTION

Graph Contrastive Learning (GCL) has become a cornerstone of modern graph representation learning, enabling models to learn powerful node embeddings without relying on manual labels [hassani_2020_contrastive, xu_2021_infogcl]. The core principle involves maximizing the similarity between different augmented ‘views’ of the same node (positive pairs) while minimizing the similarity between views of different nodes (negative pairs). While empirically successful, the field has been largely guided by the ‘Neural Collapse’ (NC) phenomenon, which observes that in the terminal phase of training, last-layer features of the same class collapse to their means, and these class means form a Simplex Equiangular Tight Frame (ETF) ?. This geometry represents a state of maximal separation and has been considered an optimal target for representation learning.

However, this paradigm is facing a significant challenge. Recent theoretical work has uncovered a ‘low-rank bias’ in deep models, demonstrating that the ETF structure is sub-optimal for multi-layer architectures ?. Instead, the inherent regularization in deep learning favors solutions with a more structured, lower-rank geometry. This finding implies that blindly pushing GCL representations towards an ETF may be counterproductive and misaligned with the natural tendencies of the models we seek to train. This theoretical shift highlights a major difficulty and limitation in current GCL methodologies. Existing regularization techniques designed to prevent representation collapse and oversmoothing are ill-suited for this new paradigm. For instance, global regularizers like ContraNorm are highly effective but impose a computational cost of $O(n^2)$, rendering them infeasible for the massive graphs prevalent in real-world applications ?. On the other hand, the standard InfoNCE loss used in most GCL frameworks is inefficient, as it expends significant model capacity repelling ‘easy’

negatives that pose no risk of collapsing, rather than focusing on the ‘hard’ negatives that truly define the representation manifold.

To address these challenges, we propose Adaptive Structure-Aware Repulsion (ASAR), a scalable, loss-based regularizer designed to actively guide GCL representations toward the theoretically motivated low-rank geometry. ASAR introduces a novel dual-adaptive mechanism that distinguishes it from prior work. First, it employs dynamic margin-based hard negative mining. For each anchor node in a mini-batch, ASAR identifies a small, ‘confusable’ set of hard negatives whose similarity violates a dynamically adjusting margin. This focuses the repulsive force only where it is needed most. Second, ASAR uses a self-organizing repulsion weight. A localized repulsion loss is applied exclusively to these confusable pairs, and its strength is controlled by a dynamic weight that acts as a feedback mechanism, automatically increasing repulsion as representations start to collapse and decreasing it as they become well-separated. This creates a self-organizing system that avoids both total collapse and excessive dispersion, promoting a structured representational space.

Our contributions can be summarized as follows:

- **Scalable Adaptive Regularizer:** We propose ASAR, a scalable $O(b^2)$ loss-based regularizer with a dual-adaptive mechanism for dynamic hard negative mining and self-organizing repulsion strength, making it suitable for large-scale graphs.
- **Operationalizing Low-Rank Bias Theory:** We provide the first practical algorithm designed to explicitly operationalize the emerging ‘low-rank bias’ theory in GCL, moving the field’s objective beyond the traditional Simplex ETF model of Neural Collapse.
- **Improved Stability and Robustness:** ASAR’s self-correcting dynamic weighting scheme improves training stability and reduces sensitivity to hyperparameter tuning, making GCL more robust and practical.
- **Comprehensive Experimental Design:** We designed a comprehensive experimental suite on large-scale OGB datasets to validate our method against strong baselines and ablation models, aiming to establish a new state-of-the-art for GCL on massive graphs. Future work could explore ASAR as a diagnostic tool for identifying structurally ambiguous communities within graphs.

2 RELATED WORK

Graph Contrastive Learning. The foundation of our work lies in GCL methods that learn representations by contrasting augmented views of a graph. Foundational methods such as GRACE and GCA established the core principles of using data augmentation to create positive pairs and contrasting them against negative samples within a batch ?. These methods typically employ a standard InfoNCE loss, which treats all negative samples with equal importance. ASAR diverges from this approach by introducing an intelligent negative sampling strategy that dynamically identifies and focuses on hard negatives, thereby making the learning process more efficient. While other works have explored the role of augmentations and negative sampling [hassani_2020_contrastive, xu_2021_infogcl], our approach is novel in its use of a dual-adaptive mechanism where both the definition of a ‘hard’ negative and the strength of the repulsion are dynamically adjusted based on the current state of the embeddings.

Oversmoothing and Regularization. A key challenge in training deep Graph Neural Networks (GNNs) is the oversmoothing phenomenon, where node representations converge to a single point, losing their discriminative power. Several methods have been proposed to mitigate this, often by adding regularization. A recent and powerful approach is ContraNorm, which introduces a normalization layer that implicitly shatters representations to encourage a more uniform distribution ?. While theoretically sound, ContraNorm’s primary limitation is its $O(n^2)$ complexity with respect to the number of nodes (n), making it computationally infeasible for the large-scale graphs we target, such as those in the Open Graph Benchmark. In contrast, ASAR provides a scalable $O(b^2)$ alternative by operating at the mini-batch level. It achieves a similar goal of preventing collapse through a loss-based mechanism rather than a global architectural modification, making it a practical solution for massive graphs.

Representation Geometry and Neural Collapse. The geometry of learned representations has been a subject of intense study. The ‘Neural Collapse’ (NC) phenomenon describes how, under ideal conditions, feature vectors for each class collapse to their mean, and these class means form a Simplex Equiangular Tight Frame (ETF) ?. This has been observed in GNNs as well, although the collapse is often partial due to the constraints of the graph structure ?. The ETF geometry has been implicitly treated as the optimal target for many contrastive learning frameworks. However, recent work challenges this notion, proposing that a ‘low-rank bias’ inherent in deep models makes the ETF structure sub-optimal ?. This new theory suggests that a more structured, lower-rank geometry is the natural and more desirable outcome of training. Our work is the first, to our knowledge, to propose a practical GCL algorithm that explicitly moves away from the ETF objective and is designed to facilitate the formation of these more complex low-rank structures. While other works have analyzed the fine-grained structure hidden within collapsed representations ?, ASAR is a constructive method that actively shapes this structure during training.

3 BACKGROUND

To understand our proposed method, it is essential to first review the foundational concepts of Graph Contrastive Learning and the evolving theoretical understanding of optimal representation geometry.

Graph Contrastive Learning (GCL). GCL is a self-supervised learning framework for graphs. Given a graph $G = (V, E)$ with node features X , the goal is to learn a GNN-based encoder f that maps each node v_i to a low-dimensional representation $z_i = f(X, A)$, where A is the adjacency matrix. GCL operates by creating two correlated ‘views’ of the input graph, typically through stochastic data augmentations like node feature masking or edge dropping. For a given node v_i , its representations in the two views, z_i and z_i^+ , form a ‘positive pair’. Representations of any two different nodes, (z_i, z_j) where $i \neq j$, form a ‘negative pair’. The encoder is trained to maximize the similarity of positive pairs while minimizing the similarity of negative pairs. This is commonly achieved using a normalized temperature-scaled cross-entropy loss (InfoNCE), which for a positive pair (z_i, z_i^+) within a mini-batch B is formulated as:

$$L_i = -\log \frac{\exp(\text{sim}(z_i, z_i^+)/\tau)}{\exp(\text{sim}(z_i, z_i^+)/\tau) + \sum_{j \neq i} \exp(\text{sim}(z_i, z_j)/\tau)} \quad (1)$$

Here, $\text{sim}(u, v)$ is a similarity function (e.g., cosine similarity) and τ is a temperature hyperparameter.

Problem Setting and Notation. We consider the task of unsupervised node representation learning on a graph G . The GNN encoder f_θ maps input node features and graph structure to a d -dimensional embedding space. During training, we sample mini-batches of nodes B . For each node $i \in B$, we generate two augmented views, resulting in embeddings z_i and z_i^+ . Our goal is to learn the parameters θ of the encoder such that the resulting embeddings are highly informative for downstream tasks, such as node classification, while exhibiting a desirable geometric structure that prevents dimensional collapse.

From Neural Collapse to Low-Rank Bias. A key theoretical concept motivating our work is the geometry of learned features. The theory of Neural Collapse (NC) posits that for a perfectly balanced classification task trained to convergence, the learned features exhibit a Simplex Equiangular Tight Frame (ETF) structure ?. This means within-class variability collapses to zero, and the class means are maximally and equally separated. This has been a guiding principle for understanding and improving representation quality ?. However, a recent theoretical breakthrough has shown that for deep, non-linear models, the ETF geometry is actually sub-optimal ?. This ‘low-rank bias’ theory demonstrates that the implicit regularization of deep architectures favors solutions of an even lower rank than the ETF. This suggests that the optimal representation geometry is more structured than a simple simplex, and methods should encourage this structure rather than fight it. ASAR is designed to operate within this new paradigm, using adaptive repulsion to prevent a full, unstructured collapse while allowing for the formation of a more nuanced, low-rank representation space.

4 METHOD

Our proposed method, Adaptive Structure-Aware Repulsion (ASAR), is a loss-based regularizer designed to overcome the limitations of existing Graph Contrastive Learning (GCL) techniques. It is

scalable, efficient, and theoretically grounded in the emerging 'low-rank bias' paradigm ?. ASAR introduces a dual-adaptive mechanism that intelligently manages repulsive forces within a mini-batch to prevent representation collapse while promoting a well-structured embedding space. The total loss function is a combination of the standard alignment loss for positive pairs and our adaptive repulsion loss: $L = L_{\text{align}} + \lambda_t \cdot L_{\text{ASAR}}$.

4.1 DYNAMIC MARGIN-BASED HARD NEGATIVE MINING

A major inefficiency in standard contrastive learning is the indiscriminate repulsion of all negative pairs. Most negative samples are 'easy' and already far apart in the embedding space, and applying repulsive force to them wastes model capacity. ASAR focuses only on the most challenging or 'confusable' negative pairs that risk collapsing the representation manifold. For each anchor embedding z_i in a mini-batch B , we define a set of confusable hard negatives $C(i)$. A negative sample z_j is considered confusable if its similarity to z_i violates a dynamic margin δ_t relative to the positive pair's similarity:

$$C(i) = \{z_j \mid j \neq i \text{ and } \text{sim}(z_i, z_j) > \text{sim}(z_i, z_i^+) - \delta_t\} \quad (2)$$

Critically, the margin δ_t is not a fixed hyperparameter but adapts to the current geometry of the batch embeddings. We define it as being proportional to the diversity of similarities within the batch, for instance:

$$\delta_t = \alpha \cdot \text{stddev}(\{\text{sim}(z_k, z_l) \mid k, l \in B, k \neq l\}) \quad (3)$$

where α is a sensitivity hyperparameter. This dynamic margin ensures that as the representations become more collapsed (i.e., lower standard deviation of similarities), the criterion for being a 'hard negative' becomes stricter, focusing the model's effort on breaking up the most closely clustered, ambiguous node pairs.

4.2 LOCALIZED REPULSION WITH SELF-ORGANIZING WEIGHTING

Once the set of confusable pairs $C(i)$ has been identified for each anchor, ASAR applies a targeted repulsive loss, L_{ASAR} , exclusively to these pairs. The loss is formulated as the sum of exponentiated similarities for all identified hard negative pairs:

$$L_{\text{ASAR}} = \frac{1}{|B'|} \sum_{i \in B'} \left[\sum_{j \in C(i)} \exp(\text{sim}(z_i, z_j)/\tau) \right] \quad (4)$$

where τ is the temperature parameter and B' is the set of anchors with at least one confusable negative. This localized application ensures that no computational effort is wasted on easy negatives.

The second key component of our method is the dynamic repulsion strength, λ_t . Instead of being a fixed hyperparameter requiring extensive tuning, λ_t is a self-organizing weight that functions as a feedback controller. It is adjusted based on the batch's average pairwise similarity, sim_{avg} . The formulation is:

$$\lambda_t = \lambda_0 \cdot f(\text{sim}_{\text{avg}}) \quad (5)$$

where λ_0 is a base scaling factor and f is a modulating function, such as a clipped ramp function $f(x) = \text{clamp}(x + c, \min = 0)$. As representations begin to collapse, sim_{avg} increases, which in turn increases λ_t , strengthening the repulsive force to counteract the collapse. Conversely, as representations spread out, sim_{avg} decreases, and λ_t is reduced to prevent excessive dispersion. This creates a self-regulating system that maintains a balanced and structured embedding space, improves training stability, and significantly reduces hyperparameter sensitivity. The overall complexity remains $O(b^2)$ per batch, making ASAR a scalable and practical solution for large graphs.

5 EXPERIMENTAL SETUP

To empirically validate the performance and scalability of our proposed Adaptive Structure-Aware Repulsion (ASAR) method, we designed a comprehensive set of experiments on large-scale graph datasets.

Objective and Hypotheses. Our primary objective is to demonstrate that ASAR achieves state-of-the-art performance in downstream node classification while being computationally tractable for

massive graphs. We hypothesize that: (1) The full ASAR model will outperform standard GCL baselines and its own ablated variants in terms of classification accuracy. (2) ASAR’s training time and memory usage will be comparable to other $O(b^2)$ methods, confirming its scalability.

Datasets. We selected two widely-used large-scale benchmarks from the Open Graph Benchmark (OGB) suite. First, `ogbn-arxiv`, a citation network with approximately 170,000 nodes and 1.2 million edges, where the task is to predict paper subject areas. Second, `ogbn-products`, a much larger Amazon co-purchase network with over 2.4 million nodes and 61 million edges, serving as a stress test for scalability. For preprocessing, we followed standard OGB protocols, including the addition of self-loops.

Models for Comparison. Our evaluation includes several models:

- **Baselines:** We compare against GRACE, a foundational GCL method, and GCA, an enhanced version with adaptive augmentations ?.
- **Proposed Model:** The full ASAR model with both the dynamic hardness margin (δ_t) and the self-organizing repulsion weight (λ_t).
- **Ablation Studies:** To isolate the impact of each component, we include two variants: ASAR (Fixed Repulsion Weight), where λ_t is replaced with a fixed, tuned hyperparameter λ , and ASAR (Fixed Hardness Margin), where δ_t is replaced with a fixed hyperparameter δ .
- **Theoretical Baseline:** We note that ContraNorm ?, a global $O(N^2)$ regularizer, is computationally ‘Infeasible’ for these datasets on modern hardware, highlighting the need for scalable solutions like ASAR.

Implementation Details. All experiments were designed to be run on an NVIDIA A100 GPU with 40GB of VRAM, using PyTorch, PyTorch Geometric, and the OGB library. Due to the dataset sizes, we use the GraphSAINT sampler for mini-batching, with a batch size of 2048 nodes. The backbone encoder for all models is a standard two-layer Graph Convolutional Network (GCN) with a hidden dimension of 256, followed by a 2-layer MLP projection head (256 $\rightarrow$ 128) used during pre-training. The AdamW optimizer is used with a learning rate of 1e-3 and a cosine annealing scheduler. The temperature τ is fixed at 0.1 for all contrastive calculations.

Evaluation Protocol. The models undergo unsupervised pre-training for 100 epochs. Following this phase, the GCN encoder’s weights are frozen, and a logistic regression classifier is trained on the node embeddings to evaluate their quality on the downstream node classification task. This linear probing protocol ensures a fair comparison of the learned representations. To ensure statistical reliability, we planned to repeat each experiment 5 times with different random seeds and report the mean and standard deviation of the test accuracy.

Metrics. We measure three key metrics: (1) Test accuracy for the downstream node classification task. (2) Total wall-clock time for the 100-epoch pre-training phase. (3) Peak GPU memory consumption during pre-training.

6 RESULTS

We executed the experimental protocol described in the previous section to evaluate our proposed ASAR method. However, the experiments could not be completed due to critical failures during the initial data loading stage, preventing any model training or result generation.

Experimental Execution and Failures. The experimental script was run in an environment with PyTorch, PyTorch Geometric, and OGB libraries installed on a CUDA-enabled device. The process halted prematurely while attempting to load the required datasets. The logs reported two distinct errors:

1. **ogbn-arxiv Loading Failure:** When attempting to load the `ogbn-arxiv` dataset, the process terminated with a `Weights only load failed` error. The error message indicated this is likely due to a version incompatibility between the PyTorch environment (≥ 2.6) and the format of the saved dataset file. The default security settings in recent PyTorch versions for `torch.load` prevented the data from being unpickled.

2. **ogbn-products Loading Failure:** Following the first failure, the script attempted to load the ogbn-products dataset. This also failed, but with an EOF when reading a line error. The log notes suggest this type of error can be expected on systems with limited hardware or network resources, possibly indicating an incomplete download or file corruption of the large (1.38 GB) dataset.

A summary of the execution log, which captures these failures, is presented in Figure 1.

```
{
  "experiment_status": "FAILED",
  "error_summary": {
    "ogbn-arxiv": "Weights only load failed. torch.serialization.add_safe_global.",
    "ogbn-products": "EOF when reading a line. Potentially incomplete download or",
  },
  "models_trained": [],
  "results_generated": false
}
```

Figure 1: Summary log of the experimental execution, indicating data loading failures. (filename: experiment_summary.json)

Discussion and Limitations. As a direct consequence of these data loading failures, no models were pre-trained, and no performance metrics (accuracy, training time, peak memory) were collected. Therefore, we are unable to present any empirical results to validate our hypotheses regarding the performance of ASAR compared to baselines or its ablated variants. The primary limitation of this work is the lack of empirical validation for the proposed method. The experimental design remains robust, but its execution was thwarted by technical and environmental issues that must be resolved before any conclusions about ASAR’s practical efficacy can be drawn. This outcome underscores the practical challenges of working with large-scale public datasets and the importance of maintaining compatible software environments for reproducibility.

7 CONCLUSION

In this paper, we addressed a critical gap between the theoretical foundations and practical implementations of Graph Contrastive Learning (GCL). Motivated by the recent ‘low-rank bias’ theory ?, which challenges the optimality of the classic Neural Collapse paradigm ?, we identified the need for a scalable regularizer that can guide representations toward a more structured, theoretically-sound geometry.

Our primary contribution is the proposal of Adaptive Structure-Aware Repulsion (ASAR), a novel loss-based regularizer with a dual-adaptive mechanism. ASAR efficiently focuses repulsive forces on ‘confusable’ hard negatives identified via a dynamic margin and modulates the repulsion strength using a self-organizing weight that acts as a feedback controller. With its $O(b^2)$ complexity, ASAR is designed to be a practical and powerful tool for training GNNs on massive graphs, where existing $O(n^2)$ methods are infeasible.

We designed a thorough experimental protocol to validate ASAR’s performance on large-scale OGB benchmarks. However, due to unforeseen technical issues during the data loading phase, we were unable to complete the experiments and gather empirical evidence. Consequently, the practical validation of ASAR remains an important and unresolved question.

For future work, the immediate next step is to resolve the environmental and data-handling issues to execute the planned experiments and empirically verify our claims. Assuming successful validation, further research could explore the application of ASAR’s principles to other domains, such as vision and language, where contrastive learning is prevalent. Additionally, the set of ‘confusable’ nodes identified by ASAR could be leveraged as a novel diagnostic tool to analyze and uncover structurally ambiguous or challenging regions within complex graphs, opening new avenues for graph interpretation and analysis.

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REFERENCES

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